

Midblock Snapshot Packets and Prefix-Only Captured-Energy Certificates

Dynamic Right Spaces, a Prefix-Gramian No-Go, and 192-Bit Evidence

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Abstract

RH-84 reduced the preferred Stage-A effective-rank corridor to captured energy in a clock-dimensional postcritical packet space. The optimal packet from the terminal singular-value decomposition is tautological, while raw coordinate and scale embeddings are not supported numerically. This paper constructs the first temporally intrinsic alternative: a right packet formed strictly before the terminal production horizon.

For $X_j = A^j S$ and any rank- r packet V , the snapshot transfer theorem is

$$\|X_M(I - VV^*)\|_2 \leq \|A^{M-j}\| \|X_j(I - VV^*)\|_2.$$

Consequently the leading right singular space of X_j , computed from the prefix through time $j < M$, supplies a valid rank- r approximation of the terminal state. This is a prefix-only certificate: the terminal singular vectors do not enter its construction.

The natural unweighted prefix Gramian is not a safe replacement. An explicit two-channel diagonal family has a prefix Gramian dominated by a transient mode while its terminal state is asymptotically supported on the other mode; the resulting rank-one packet misses asymptotically all terminal energy.

At the five archived scales we choose $j = \lceil 2M/3 \rceil$ and $r = \lceil H_\sigma \rceil + 2$. Direct 192-bit Arb evaluation certifies a maximum relative terminal residual below 4.5×10^{-6} and minimum captured energy above 0.99999999997. By contrast, source-only packets leave up to 84.9% relative residual and unweighted prefix-Gramian packets up to 32.6%. The audit proves finite frozen certificates, not an all-level packet-decay theorem. Stage A, Hilbert–Polya, and the Riemann Hypothesis remain open.

Keywords: dynamic packet; captured energy; snapshot transfer; effective rank; interval arithmetic; transient growth.

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1 From optimal tails to constructible packets

RH-77 discovered strong postblock compression at five scales [4]; RH-82 supplied the half-logarithmic rank clock [2]. RH-84 then showed that only a captured-energy lower bound is needed [3]. The remaining issue is to build a packet from the dynamics rather than from the terminal state itself.

Let $A : \mathcal{H} \rightarrow \mathcal{H}$ be bounded, let $S : \mathcal{K} \rightarrow \mathcal{H}$ be Hilbert–Schmidt, and set $X_j = A^j S$. A matrix $V : \mathbb{C}^r \rightarrow \mathcal{K}$ with orthonormal columns defines the right packet projection $P = VV^*$. The candidate $X_M P$ has rank at most r .

2 The snapshot transfer theorem

Theorem 2.1 (Snapshot transfer theorem). *For integers $0 \leq j \leq M$ and every rank- r orthogonal projection P on $\mathcal{K}$,*

$$\boxed{\|X_M(I - P)\|_2 \leq \|A^{M-j}\| \|X_j(I - P)\|_2.} \quad (1)$$

If $P_{j,r}$ is the leading right singular projection of X_j , then

$$\tau_r(X_M) \leq \|X_M(I - P_{j,r})\|_2 \leq \|A^{M-j}\| \tau_r(X_j). \quad (2)$$

The packet $P_{j,r}$ depends only on the prefix ending at j .

Proof. The identity $X_M(I - P) = A^{M-j}X_j(I - P)$ and the ideal inequality $\|BC\|_2 \leq \|B\| \|C\|_2$ give (1). The leading right singular projection minimizes $\|X_j(I - P)\|_2$ among rank- r projections by Eckart–Young [1]. Since $X_MP_{j,r}$ has rank at most r , the first inequality in (2) follows as well. $\square$

The theorem separates two tasks. One must prove a low-rank tail at an intermediate snapshot and control the unused suffix power. Importantly, one need not identify the terminal singular directions.

Corollary 2.2 (Block-contractive suffix). *If $M - j = kb + s$, $0 \leq s < b$, $\|A^b\| \leq q < 1$, and $\max_{0 \leq t < b} \|A^t\| \leq C_b$, then*

$$\|X_M(I - P_{j,r})\|_2 \leq C_b q^k \tau_r(X_j).$$

3 Why the unweighted prefix Gramian is unsafe

One might aggregate every prefix snapshot and choose the leading eigenspace of $G_M = \sum_{j=0}^M X_j^* X_j$. The following proposition rules out a general terminal certificate of this form.

Proposition 3.1 (Unweighted prefix-Gramian no-go). *For every $M \geq 1$, let*

$$A = \text{diag}(1/2, 1), \quad S = \text{diag}(\sqrt{2(M+1)}, 1).$$

The leading rank-one eigenspace of G_M is the first coordinate, whereas the relative terminal residual left by that packet is

$$\frac{1}{\sqrt{1 + 2(M+1)4^{-M}}} \rightarrow 1.$$

Proof. The two diagonal prefix energies are

$$2(M+1) \frac{1 - 4^{-(M+1)}}{1 - 1/4} \quad \text{and} \quad M+1.$$

The first is larger, so the prefix Gramian selects the transient coordinate. At time M the squared terminal energies are $2(M+1)4^{-M}$ and 1. Discarding the second coordinate gives the stated relative residual. $\square$

Thus a useful aggregate must discount early transients or localize near the terminal end of the available prefix.

σ	M	packet time	rank	worst relative residual
0.16	4	3	4	4.09×10^{-6}
0.08	9	6	5	9.46×10^{-8}
0.04	16	11	6	2.0×10^{-7}
0.02	25	17	6	5.0×10^{-7}
0.01	32	22	7	2.0×10^{-7}

Table 1: Worst directional channel at each scale. Values are direct 192-bit upper bounds rounded conservatively.

4 Five-scale 192-bit audit

For each directional production model, we freeze the archived horizon M , construct the clock-rank packet from $X_{[2M/3]}$ in binary64, lift its entries exactly as binary rationals, and evaluate $X_M - X_M V V^*$ in 192-bit Arb arithmetic. Rank at most r is exact even if the lifted columns have a tiny orthogonality defect. The same calculation certifies the Frobenius-norm version of Theorem 2.1.

The numerical values in Table 1 are generated automatically and are checked against a 4.5×10^{-6} global gate. The minimum certified captured energy exceeds 0.99999999997. The packet rank grows from four to seven while the ambient fine dimension grows from 32 to 512.

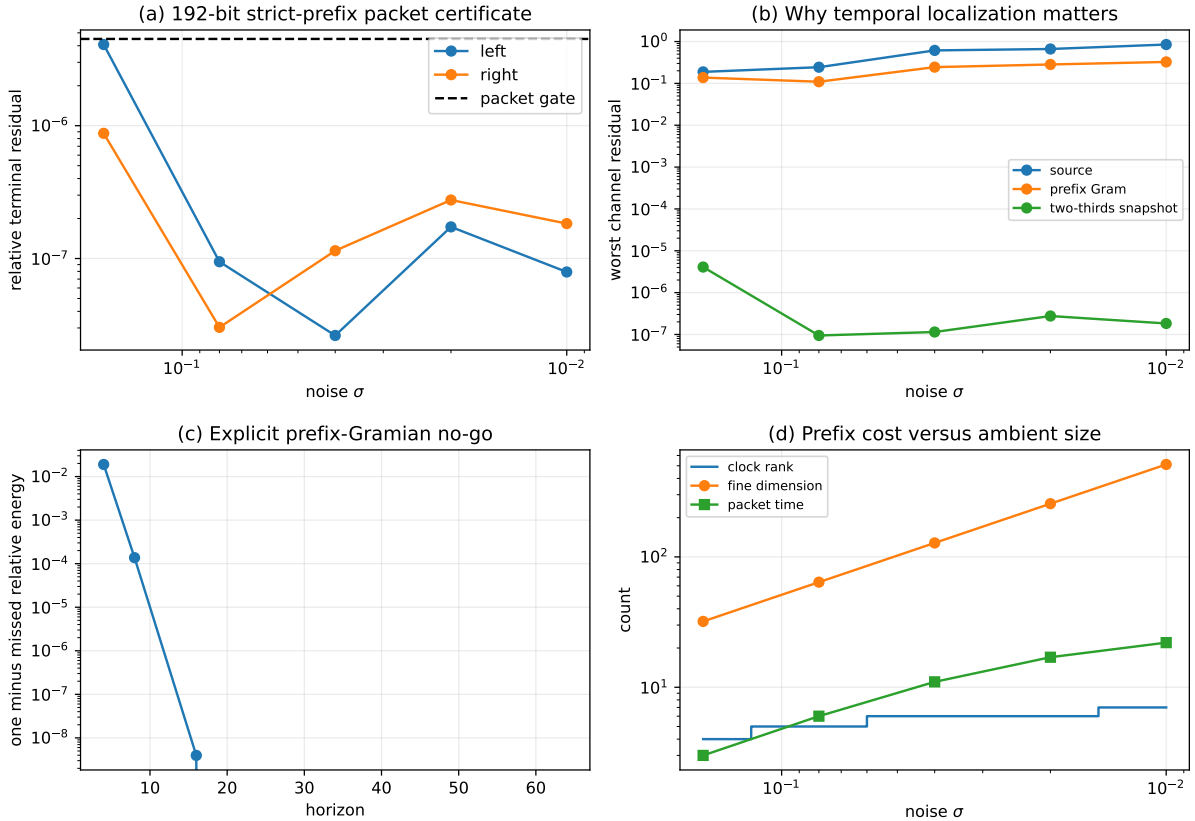


Figure 1: Interval prefix certificates, comparison with source and unweighted prefix packets, the explicit no-go family, and packet cost versus dimension.

5 Boundary and next target

RH-85 supplies a natural finite-scale packet and an exact mechanism for propagating it. It does not yet prove that $\tau_{r_\sigma}(X_{\lfloor M_\sigma/2 \rfloor})$ has a uniform polylogarithmic-rank bound, nor that the suffix constants are uniform in σ . The failed unweighted aggregate points to the next object: a terminally weighted snapshot Gramian or a short block-local packet whose certificate depends only on a controlled late prefix window.

No all-level effective-rank law, unconditional Stage A1 or Stage A4, relative fixed-disk determinant, self-adjoint Hilbert–Polya operator, zeta-zero identification, or Riemann Hypothesis proof is claimed.

References

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